

Date: ____/____/____

MON—TUE—WED—THUR—FRI—SAT—SUN

Q1) $f: A \rightarrow B$

$C \subseteq A$

$$f|_C = f \cap (C \times B)$$

(a)

$f|_C$ is a function: $C \rightarrow B$

$$\forall c \in C \exists ! b \in B (c, b) \in f|_C$$

let $c \in C$ arbitrary.

$$\exists ! b \in B (c, b) \in f \cap (C \times B)$$

~~Supp~~ Existence: Note that $c \in A$ as well

let $b = f(c)$ so $b \in B$. so $(c, b) \in C \times B$ & as $C \subseteq A$ so

$c \in A$ so $(c, b) \in f$. so $(c, b) \in f \cap (C \times B)$

Uniqueness:

$$\forall b_1 \in B [(c, b_1) \in f|_C \rightarrow b_1 = b]$$

~~Supp~~ $b_1 \in B$

$$\& (c, b_1) \in f|_C$$

$$b_1 = b$$

so $(c, b_1) \in f \cap (c, b_1) \in C \times B$

but we have $(c, b) \in f$ & f is a function so it must be that $b = b_1$.

To derive formula $f|_C(c) = (f|_C)(c)$

~~(f|_C)(c)~~

~~Supp~~ $c \in C$

$\rightarrow$ ~~Supp~~ $f(c)$

$$\rightarrow (f|_C)(c)$$

let $b = f(c) \in B$

$$f \cap (C \times B)(c)$$

so $(c, b) \in f$

$$(c, b) \in f \cap (C \times B)$$

Clearly $c \in C$ & $b \in B$ so $(c, b) \in C \times B$ & we already have

that $(c, b) \in f$. since we already proved it's a function

$\leftarrow$ ~~Supp~~ $(c, b) \in f|_C$. Clearly $(c, b) \in f$ as $f|_C$

$$= f \cap (C \times B).$$

Mergestack

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Date: / /

MON - TUE - WED - THUR - FRI - SAT - SUN

(b) $g: C \rightarrow B$ $g = f \circ c \iff g \leq f$
 $\rightarrow$ $\text{supp } g = f \circ c$ $g \leq f$
 $\text{Supp } (c, b) \in g$ be arbitrary $(c, b) \in f$
 $\Rightarrow g = f \circ c$ so $(c, b) \in f \circ c$ so $(c, b) \in f \cap (C \times B)$
 so $(c, b) \in f$.
 $\leftarrow$ $\text{supp } g \leq f$ $g = f \circ c$
 Any arbitrary $(c, b) \in g$ iff $(c, b) \in (C \times B) \cap (c, b) \in f \circ c$
 iff $(c, b) \in f \cap (C \times B)$ iff $(c, b) \in f$

(c) $g = \{(x, y) \in \mathbb{Z} \times \mathbb{R} \mid y = 2x + 3\}$
 $h = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y = 2x + 3\}$
 $h(x)$

$g = h \circ 2$
 $\rightarrow \text{supp } (x, y) \in g$ $(x, y) = h \circ 2$
 so $(x, y) \in \mathbb{Z} \times \mathbb{R} \wedge y = 2x + 3$ $(x, y) \in h \cap (\mathbb{Z} \times \mathbb{R})$
 $\Rightarrow$ so that $(x, y) \in h$
 as $\mathbb{Z} \subseteq \mathbb{R}$ so $x \in \mathbb{R}$ as well so $(x, y) \in \mathbb{R} \times \mathbb{R}$ as well
 $\in y = 2x + 3$ as we already know
 $\leftarrow \text{Supp } (x, y) \in h \circ 2$ $(x, y) \in g$
 iff $(x, y) \in \mathbb{R} \times \mathbb{R} \wedge (x, y) \in 2 \times \mathbb{R} \wedge y = 2x + 3$
 clearly goal follows.

$y = 2x + 3$

Mergestack

Date: / /

MON - TUE - WED - THUR - FRI - SAT - SUN

(Q8) $f: A \rightarrow B$ $\exists A' \subseteq A (g: A' \rightarrow B)$
 $g \subseteq f$ $g \subseteq B$
 let $A' = \{a \mid (a, b) \in f\} = \text{Domain of } f$
 let $a \in A'$ be arbitrary as $(a, b) \in f \Rightarrow g \subseteq f$
 so $(a, b) \in f$ so $a \in A$ so $A' \subseteq A$.
 To see that g is a function $\forall a \in A' \exists! b \in B (a, b) \in g$
 let $a \in A'$ $A' \subseteq A$ so $a \in A$ as well
 let $b = f(a) \in B$ so $(a, b) \in f$
 then by (i) we defined there must some $b \in B$ with $(a, b) \in f$
 Uniqueness
 $\text{supp } (a, b_1) \in g \wedge (a, b_2) \in g$ $b_1 = b_2$
 so $a \in A'$ $b_1, b_2 \in B$ $\therefore g \subseteq f$
 as $A' \subseteq A$ so $a \in A$
 so $(a, b_1) \wedge (a, b_2) \in f$ but f is a function so $b_1 = b_2$

(Q9) $y: A \rightarrow B$
 $B \neq \emptyset$ $\exists g: A' \rightarrow B (g \subseteq y)$
 $A \subseteq A'$
 By analysing an example we can see that we don't want to repeat the domain. $2 B \neq \emptyset$ so let $b \in B$ - let this be only element that we will map elements of A' so domain won't be repeated. Also g should contain f . So g is analysing an example: $g = f \cup \{(A' \setminus A) \times \{b\}\}$ looks like right choice

Mergestack

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

we will prove first that $g: A' \rightarrow B$
 existence: supp $a \in A'$. Then there is $b_0 \in B$ such
 that by defn of g , $(a, b_0) \in g$.

Uniqueness:

supp $(a, b_1) \in g \wedge (a, b_2) \in g$. $b_1 = b_2$
 $\because$ As $g = f \cup \{(A' \setminus A) \times \{b_0\}\}$. We consider 2 cases

case 1) supp $(a, b_1) \notin f$.

case 1.1) $(a, b_2) \in f$. The f is fctn so both

case 1.2) $(a, b_2) \in \{(A' \setminus A) \times \{b_0\}\}$ the
 $a \in A' \setminus A$ which is contradiction over $a \in A$

as $(a, b_1) \in f$. So this case is not possible.

case 2) $(a, b_1) \in \{(A' \setminus A) \times \{b_0\}\}$. Then $b_1 = b_0$

if $(a, b_2) \in f$. The $a \in A$ which is contradiction
 over $a \notin A$. So this is not possible.

so $(a, b_2) \in \{(A' \setminus A) \times \{b_0\}\}$ so $b_2 = b_0$

so $b_1 = b_2 = b_0$.

therefore proved.

Finally to see that $f \subseteq g$.

supp $(a, b) \in f$. then $\rightarrow (a, b) \in g$.
 as $g = f \cup \{(A' \setminus A) \times \{b_0\}\}$ so $(a, b) \in g$.

Q10) $f: A \rightarrow B, g: A \rightarrow B, f \neq g$.

$f \circ g$ is not a fctn on
 $= (f \cup g) \setminus (f \cap g)$ is not a fctn

Mergestack

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

By translating logical symbols, we can see that as $f \neq g$
 we can choose some $(a, b) \in f \setminus g$ such that $(a, b) \notin f \cap g$
 $\forall (a, b) \in g \cap f \setminus b \notin f$.

Case 1) $(a, b) \in f \setminus (a, b) \in g$.

We will use more that $f = g$ iff $\forall a \in A (f(a) = g(a))$
 as $f \neq g$. so we can choose

some $a_0 \in A (f(a_0) \neq g(a_0))$

Given $f: A \rightarrow B$ so $(a_0, f(a_0)) \in f$. as g is a fctn.

so similarly $(a_0, g(a_0)) \in g$ so $(a_0, f(a_0)) \in f \setminus g$

then $f(a_0) \neq g(a_0)$ so similar argument
 $f \circ g$ is not a fctn.

Q11) we have already seen that it is equivalent in Q1.4 &
 we saw in Q1.5 it's a fctn.

Uniqueness:

supp f & g are 2 fctns on A that are also equivalent
 relations $\rightarrow f = g$

so $f: A \rightarrow A, g: A \rightarrow A$

$\forall a \in A (f(a) = g(a))$

let $a \in A$.

By reflexivity of f & g , $(a, a) \in f \cap (a, a) \in g$

so $f(a) = a = g(a)$ Hence proved

Q12)

Mergestack

Date: ____/____/____

$$f: A \rightarrow C$$

MON - TUE - WED - THUR - FRI - SAT - SUN

$$Q12) (b) \rightarrow \text{supp } f \cup \text{supp } g: A \cup B \rightarrow C$$

$$\begin{aligned} & \text{supp } f \cup \text{supp } g: A \cup B \rightarrow C \\ & \text{supp } f \cup \text{supp } g: A \cup B \rightarrow C \\ & \text{supp } f \cup \text{supp } g: A \cup B \rightarrow C \\ & \text{supp } f \cup \text{supp } g: A \cup B \rightarrow C \end{aligned}$$

$$\begin{aligned} & \text{so } \exists y_1 \in C \text{ s.t. } (x, y_1) \in f \\ & \text{if we can prove } y_1 = y_2 \text{ then we are done} \\ & \text{as } x \in A \cap B \text{ so } x \in A \text{ and } x \in B \\ & \text{so } \exists y_1 \in C \text{ s.t. } (x, y_1) \in f \text{ and } (x, y_2) \in g \\ & \text{case 1) } (x, y_1) \in f \text{ then } y_1 = y_2 \text{ (if } f \text{ is } f \text{ then)} \\ & \text{case 2) } (x, y_2) \in g \text{ then } y_1 = y_2 \end{aligned}$$

$$\text{so } y_1 = y_2$$

$$\text{supp } f \cup \text{supp } g: A \cup B \rightarrow C$$

$$\text{so } \exists c_1 \in C \text{ s.t. } (a, c_1) \in f \text{ and } (a, c_2) \in g$$

$$\text{so } c_1 = c_2$$

$$\text{Note: The problem was that I was considering cases that } a \in A$$

$$\text{so } a \in A \cup B \text{ then there are 2 cases, } a \in A \text{ or } a \in B \text{ but}$$

$$\text{we should also try } f \cup g \text{ as well. } a \in A \cap B \text{ so using only}$$

$$\text{is not as powerful as using } f \cup g$$

Mergestack

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

$$\begin{aligned} & \text{supp } f \cup \text{supp } g: A \cup B \rightarrow C \\ & \text{supp } f \cup \text{supp } g: A \cup B \rightarrow C \\ & \text{supp } f \cup \text{supp } g: A \cup B \rightarrow C \\ & \text{supp } f \cup \text{supp } g: A \cup B \rightarrow C \end{aligned}$$

$$\text{Uniqueness:}$$

$$\text{supp } f \cup \text{supp } g: A \cup B \rightarrow C$$

$$\text{so } c_1 = c_2$$

$$\text{Q13) } R \subseteq A \times B, S \subseteq B \times C$$

$$\text{Ker}(K): \text{Dom}(S) \rightarrow B$$

$$\text{so } K: A \rightarrow C$$

$$\text{supp } K \subseteq B$$

$$\text{Uniqueness: } \text{supp } K \subseteq B$$

$$\text{so } c_1 = c_2$$

Mergestack

5 will remain a fiction

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

- (b) $R: A \rightarrow B$ $A = \{1, 2, 3\}$, $B = \{1, 2, 3\}$, $C = \{1, 2\}$
 $R = \{(1, 1), (1, 2), (2, 1)\}$
 $S = \{(1, 2), (2, 2)\}$
 $\text{so } R = \{(1, 1), (2, 2)\}$

Q13) Note: The uniqueness proof fiction is kind of proof by contradiction in which we contradict the fact that S is a relation (e.g. f) so $(a, b), (a, b_2)$ would allow to f so $b = b_2$ is must. Although it's logic of $\exists!$ at backed. That says if 2 elements belong, satisfy same property, they must be equal.

- Q14) $f: A \rightarrow B$ $S \subseteq B \times B$
 $R = \{(x, y) \in A \times A \mid (f(x), f(y)) \in S\}$

(a) S is reflexive $\rightarrow R$ is reflexive
 $\forall b \in B (b, b) \in S \rightarrow \forall a \in A (a, a) \in R$
 Suppose $a \in A$ then $\rightarrow (a, a) \in R$
 $\exists! b \in B (a, b) \in f (f(a) = b) = (f(a), f(a)) \in S$
~~then~~ $f(a) = b \in B$
 As S is ref, $(b, b) \in S$ or $(f(a), f(a)) \in S$

(b) S is symmetric $\rightarrow R$ is symmetric
 Let $a_1, a_2 \in R \wedge a_1 R a_2 \rightarrow a_2 R a_1$
 $a_1 R a_2 \rightarrow (f(a_1), f(a_2)) \in S$

Mergestack

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

R is reflexive $\rightarrow f(a) \in f(a)$ so by def of R , $a R a$.

(c) S is transitive $\rightarrow R$ is transitive
 Suppose $a_1, a_2, a_3 \in A \wedge a_1 R a_2 \wedge a_2 R a_3 \rightarrow a_1 R a_3$
 $f(a_1) \in f(a_2) \wedge f(a_2) \in f(a_3)$ by transitivity of $S \rightarrow f(a_1) \in f(a_3)$
 $\rightarrow a_1 R a_3$.

Q15) $f: A \rightarrow B$, $R \subseteq A \times A$
 $S = \{(x, y) \in B \times B \mid \exists u \in A \wedge (f(u) = x \wedge f(v) = y \wedge (u, v) \in R)\}$

(a) Already solved in week. $\rightarrow$ For S to be ref, $\forall b \in B (b, b) \in S$
 Note all elements of B (x.b)
 (c) We will help to see that when supp to S b_1, b_2, b_3
 then $f(a_1) = b_1 \in f(a_2) = b_2$ but for b_2, b_3 , then
 can be $f(a') = b_2$ (not need b_3 again)

Q16) $R \subseteq B \times B$ and sets $F = \{f \mid f: A \rightarrow B\}$
 $R \subseteq B \times B$
 $S = \{(f, g) \in F \times F \mid \forall x \in A (f(x), g(x)) \in R\}$

(a) Yes. Suppose $f \in F \rightarrow (f, f) \in S$.
 Let $x \in A$. As $f: A \rightarrow B$ so $\exists! b \in B (f(x) = b)$. As R is
 ref, so $(b, b) \in R = (f(x), f(x)) \in R$ so $(f, f) \in S$.
 (b) Yes. Suppose $(f, g) \in S$. then $\forall x (f(x), g(x)) \in R$

Mergestack

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

but $x \in A \rightarrow (g, f) \in R$
 As $x \in A$, so $(f(x), g(x)) \in R$ & R is symmetric, so $(g(x), f(x)) \in R$
 so $(g, f) \in S$.

(c) Yes. Easy proof. Similar to symmetry.

Q17) $A \neq \emptyset$, $f: A \rightarrow A$
 $\exists a \in A \forall x \in A (f(x) = a) \rightarrow \forall g: A \rightarrow A (f \circ g = f)$
 Let $g: A \rightarrow A$ be arbitrary
 $\rightarrow$ Supp $(a, a) \in f \circ g$. The $(f, x) \in g \cap (a, a) \in f$. By (1),
 $\forall x \in A (f(x) = a)$ but also $f(x) = a$, so $a = a$
 so $\forall x \in A (f(x) = a)$. Since $a \in A$, $(f, a) \in f$ so $(a, a) \in f$.
 $\rightarrow$ Supp $(f, a) = a$ as $(a, a) \in f$. As $\forall x \in A (f(x) = a)$, we
 know $a \in A$ so $f(a) = a$. So $a = a$ & $\forall x \in A (f(x) = a)$

Stuck: Do not forget that to prove that 2 functions are equal, use the definition $f = g$ iff $\forall x (f(x) = g(x))$
 So to prove this we know supp $g: A \rightarrow A$
 Now prove $\forall x \in A (f \circ g(x) = f(x))$
 Let $x \in A$. $f \circ g(x) = f(g(x))$ as $g: A \rightarrow A$
 $\exists a_1 \in A (g(x) = a_1)$ so $f(g(x)) = f(a_1)$ & as
 $f: A \rightarrow A \exists a \in A (f(a) = a)$. By the given (1),
 we can conclude $\forall g \in A (f(g) = a)$ so as $x \in A$
 $f(x) = a$ as well so $f \circ g(x) = f(g(x)) = f(a) = a = f(x)$
 so $f \circ g(x) = f(x)$

Mergestack

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

(b) $\forall g: A \rightarrow A (f \circ g = f) \rightarrow f$ is constant
 $\exists a \in A \forall x \in A (f(x) = a)$

$\rightarrow \forall g: A \rightarrow A [\forall x \in A (f \circ g(x) = f(x))]$
 $\forall g: A \rightarrow A [\forall x \in A (f(g(x)) = f(x))]$

Intuition & understanding

First $A \neq \emptyset$ so select $a \in A$. An important thing is that f is a function, so select $f(a) = a$. Last important thing is that $a \in A$ to apply universal instantiation, we have to choose any g that is a function (to use this given). Our only choice is to define $\forall x \in A (g(x) = a)$ as a is only element we have here.

Note that this does make g a constant function but to apply universal instantiation, (which needs for all $g: A \rightarrow A$), we just need one g that is a function.

Q18) $R: \{b | b: \mathbb{R} \rightarrow \mathbb{R}\}$

$R = \{(f, g) \in R \times R | \exists a \in \mathbb{R} \forall x \in \mathbb{R} (f(x) = g(x))\}$

(a) $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = |x|$

$g: \mathbb{R} \rightarrow \mathbb{R} \quad g(x) = x$

$(f, g) \in R$

$f \neq g$ are already functions.

$\exists a \in \mathbb{R} \forall x \in \mathbb{R} (f(x) = g(x))$

Let $a = 0$

Then $\forall x \in \mathbb{R} \quad f(x) = |x| = x = g(x)$

$\forall f, g \in R (f, g) \in R \rightarrow (f, g) \in R$

(b) Reflexivity, symmetry

$\forall (f, g) \in R \quad (g, f) \in R$

Let $(f, g) \in R$ then $\exists a \in \mathbb{R} \forall x \in \mathbb{R} (f(x) = g(x))$

which is same as $\exists a \in \mathbb{R} \forall x \in \mathbb{R} (g(x) = f(x))$ so $(g, f) \in R$

Mergestack

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

Supremacy of Convexity

$\forall f, g \in F(\mathbb{R})$

we $f \in F$

$\exists a \in \mathbb{R} \forall x > a, f(x) = g(x)$

Clearly for all x , $f(x) = g(x)$ as f, g are functions.

so it must be true for some $x > a$

Transitivity:

$\forall f, g, h \in F [f R g \wedge g R h \rightarrow f R h]$

Let $f, g, h \in F$

$\& \text{supp } f R g \wedge g R h \rightarrow f R h$

$f R g$ means $a_1 \in \mathbb{R} \forall x > a_1, (f(x) = g(x))$

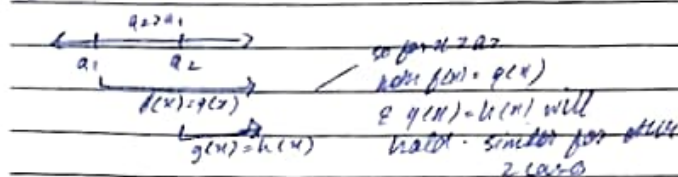
$g R h$ means $a_2 \in \mathbb{R} \forall x > a_2, (g(x) = h(x))$

We have to prove $\exists a \in \mathbb{R} \forall x > a, (f(x) = h(x))$

We can find it by looking at number line. If $a_1 = a_2$ then

$a = a_1 = a_2$ works. If $a_1 > a_2$ then $a = a_1$ works.

If $a_2 > a_1$ then $a = a_2$ works.



The reason we arrived at answer is that, if we look at g and h graphs, we have to prove $f(x) = h(x)$ & only way that looks is $f(x) = g(x)$ & $g(x) = h(x)$ so there should be some x for which both of them holds true & we can conclude $f(x) = g(x) = h(x)$.

That a might be different in case $a_1 < a_2$ or $a_1 > a_2$ or $a_1 = a_2$

Mergestack

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

(Case 1) $a_1 = a_2$. Then let $a = a_1$. Clearly $\forall x > a$,

$f(x) = g(x) = h(x)$ so $f R h$

(Case 2) $a_1 < a_2$. Then let $a = a_1$. Let $x > a_2$ be arbitrary.

As $x > a_1$ so $f(x) = g(x)$. Also $x > a_2 > a_1$ so $x > a_1$.

so $g(x) = h(x)$ & so $f(x) = h(x)$ so $f R h$

(Case 3) $a_2 < a_1$. Then let $a = a_2$. Let $x > a_1$ be arbitrary.

As $x > a_1$ so $f(x) = g(x)$. Also $x > a_1 > a_2$ so $x > a_2$.

so $g(x) = h(x)$. Therefore $f(x) = h(x)$ & thus $f R h$

Q19) (a) we just need to find one combination of a & ϵ that works. One combination of a & ϵ already implies infinite combinations as example for any $b > a$, then same ϵ would work.

$F = \{f \mid f: \mathbb{R}^+ \rightarrow \mathbb{R}\}$

for any $g \in F$, $O(g) = \{f \in F \mid \exists c \in \mathbb{R}^+ \exists a \in \mathbb{R} \forall x > a, (|f(x)| \leq |g(x)|)\}$

(b) $f(x) = 7x + 3$, $\mathbb{R}^+ \rightarrow \mathbb{R}$

$g(x) = x^2$, $\mathbb{R}^+ \rightarrow \mathbb{R}$

prove that $f \in O(g)$

$\exists a \in \mathbb{R}^+ \exists c \in \mathbb{R}^+ \forall x > a$

Let $a = 3$ & $c = 8$

Then $|f(x)| = |f(x)| = 7x + 3 \leq 7x + x = 8x \leq 8x^2 = c|g(x)|$

Basically we needed to show $\forall x > 3, |7x + 3| \leq 8x^2$

We can choose any a & ϵ that works but choose one that is easiest to prove. For a different combination, we might have to go for proof by contradiction or induction.

Mergestack

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

(b) There are 2 ways. Assume $\exists a \in \mathbb{R}^+ \exists c \in \mathbb{R}^+ \forall x > a (|f(x)| \leq c|g(x)|)$
or simply draw the negation that is
 $\forall a \in \mathbb{R}^+ \forall c \in \mathbb{R}^+ \exists x > a (|f(x)| > c|g(x)|)$

but we do second:

let $a \in \mathbb{R}^+ \& c \in \mathbb{R}^+$ $\exists x > a (x^2 > c(7x+3))$

let $x = 10c$ such that $x > a \& x > 10c$ $x^2 > c(7x+3)$

since $x > 10c$ so $x^2 > 10cx = x(7x+3x)$

also $(x+3x) > c(7x+3)$

so we find an $x > a$ that prove the negation of statement

(c) $f \in O(g)$

$f: \mathbb{R} \rightarrow \mathbb{R}$

$g: \mathbb{R} \rightarrow \mathbb{R}$

$f(x) = 5f_1(x) + t f_2(x)$

$\rightarrow f \in O(g)$

$= \exists a \in \mathbb{R}^+ \exists c \in \mathbb{R}^+ \forall x > a (|f(x)| \leq c|g(x)|) \rightarrow$

$f_1 \in O(g)$

$= \exists a_1 \in \mathbb{R}^+ \exists c_1 \in \mathbb{R}^+ \forall x > a_1 (|f_1(x)| \leq c_1|g(x)|) \rightarrow$

$f_2 \in O(g)$

$= \exists a_2 \in \mathbb{R}^+ \exists c_2 \in \mathbb{R}^+ \forall x > a_2 (|f_2(x)| \leq c_2|g(x)|) \rightarrow$

To prove the goal, we need to show that $f(x) =$

$|5f_1(x) + t f_2(x)| \leq c|g(x)|$. We want to find a

c that satisfies this property for all $x > a$. Note

we can use $|f_1(x)| \leq c_1|g(x)|$ if $x > a_1$ & similarly

Mergestack

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

for a_2 so $x > a_1 \& x > a_2$ should be checked. To check

note that $|5f_1(x) + t f_2(x)| = |5f_1(x)| + |t f_2(x)|$

Already $|f_1(x)| \leq c_1|g(x)|$ & $|f_2(x)| \leq c_2|g(x)|$

so this hints toward $c_1 + c_2$ as we $c|g(x)|$ become

$c_1|g(x)| + c_2|g(x)|$. Now only problem of s

& t remains so we can multiply with s & t also

but their absolute being they can be negative. so

$c = c_1|s| + c_2|t|$ works. Finally s & t can be 0, so

we just add 1 to make sure $c \in \mathbb{R}^+$. $c = c_1|s| + c_2|t| + 1$

For proof is simple now.

let $a = \max(a_1, a_2)$ ($\therefore$ Now $a > a_1 \& a > a_2$)

let $c = c_1|s| + c_2|t| + 1$

let $x > a$ is arbitrary then also $x > a_1 \& x > a_2$. (Now we can use $\textcircled{1}$ & $\textcircled{2}$)

then $|f(x)| = |5f_1(x) + t f_2(x)| = |5f_1(x)| + |t f_2(x)|$

$\leq c_1|s||g(x)| + c_2|t||g(x)| = (c_1|s| + c_2|t|)|g(x)|$

$\leq c|g(x)|$. ($\therefore$ as we didn't write +1 so c should be ≥ 1)

$\therefore$ we used as $|f_1(x)| \leq c_1|g(x)|$

Multiply by $|s| \rightarrow |s||f_1(x)| \leq c_1|s||g(x)|$

(Q20) $g: A \rightarrow B$

$R = \{(x, y) \in A \times A \mid g(x) = g(y)\}$

R is equivalence

Reflexivity $\forall x \in A (x, x) \in R$

Supp $x \in A$ then clearly $(x, x) \in A \times A$ & $g(x) = g(x)$

Mergestack

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

so $(x, y) \in R$.

Symmetry $\forall x \forall y \in A (xRy \rightarrow yRx)$

Supp $x \in A \wedge y \in A \wedge xRy \rightarrow yRx$

or by means $g(x) = g(y)$ which is same as $g(y) = g(x)$

so yRx .

Transitivity $\forall x \in A \forall y \in A \forall z \in A [(xRy \wedge yRz) \rightarrow xRz]$

Supp $x \in A, y \in A, z \in A$

Supp $xRy \wedge yRz \rightarrow xRz$

which means $g(x) = g(y) \wedge g(y) = g(z)$ so $g(x) = g(z)$

so xRz .

(b) R is equivalence on A

$g: A \rightarrow A/R \mid g(x) = [x]_R$

Prove that $R = \{(x, y) \in A \times A \mid g(x) = g(y)\}$

Note that R has simply those pairs (x, y) where $[x]_R = [y]_R$

$\forall x \in A \forall y \in A [(x, y) \in R \rightarrow (x, y) \in \{(x, y) \in A \times A \mid g(x) = g(y)\}]$

Supp $(x, y) \in R$

By defn of R , yRx & $x \in [x]_R$

so $y \in [x]_R$ by def of $[x]_R$ & $x \in [x]_R$

so $[x]_R = [y]_R$ by lemma 4.5.5.

so $g(x) = g(y)$ where $(x, y) \in \{(x, y) \in A \times A \mid g(x) = g(y)\}$

This is a series of equivalent statements that imply both sides.

Note that R here was like saying $R = \bigcup_{x \in A/R} x \times x$

Mergestack

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

2) Note: The compatibility relation (ex 24, sec 4.5) tells us that R is compatible with S if any 2 elts are related by R iff their elts of their same equivalence class are also related by R . So R respects the transitivity of S .

The compatibility of f means that any 2 elts who have same elts must have same f output. So f collapses every class of R to a single output.

(a) $f: A \rightarrow B$

R is equivalence relation on A

f is compatible with $R = \forall x \in A \forall y \in A (xRy \rightarrow f(x) = f(y))$

Existence: let $w = \{(x, y) \in A \times B \mid \exists z \in [x]_R (f(z) = y)\}$

Note: This can be checked by analyzing a small example. Also it's same as saying $\forall z \in [x]_R (f(z) = y)$

because f makes sure every elts in $[x]_R$ has same output in B . But mathematicians use $\exists$ to prove empty sets & any proofs.

h is a function:

$\forall [x]_R \in A/R \exists! y \in B (h([x]_R) = y)$

Let $[x]_R \in A/R$

Existence: let $y = f(x) \in B$ ($\because$ use the same x)

Clearly $x \in [x]_R$ & $f(x) = y$ & $y \in B$

Uniqueness of y : let $y_1 \in B (h([x]_R) = y_1)$ - ①

let $y_2 \in B (h([x]_R) = y_2)$ - ②

$0 \rightarrow x \in [x]_R (f(x) = y_1)$ & by def of h

Mergestack

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

$$(2) \rightarrow x_2 \in [x] \quad (f(x_2) = y_2)$$

$$1. x_1 R x \wedge x_2 R x \quad \therefore \text{By def of } [x]$$

$$\text{so } f(x_1) = f(x_2) = f(x) \quad \therefore \text{By compatibility of } f$$

$$\text{so } y_1 = y_2 \quad \therefore f(x_1) = y_1 \wedge f(x_2) = y_2$$

Uniqueness of h :

$$\text{Supp } h_1: A/R \rightarrow B \quad \text{--- (3)}$$

$$h_2: A/R \rightarrow B \quad \text{--- (4)}$$

$$\rightarrow h_1 = h_2$$

$$= \forall [x] \in A/R \quad h_1([x]) = h_2([x])$$

$$\text{Let } [x] \in A/R \quad \rightarrow h_1([x]) = h_2([x])$$

$$(3) \rightarrow ([x], y_1) \in A/R \times B \mid x_1 \in [x] \quad (f(x_1) = y_1)$$

$$(4) \rightarrow ([x], y_2) \in A/R \times B \mid x_2 \in [x] \quad (f(x_2) = y_2)$$

By def of h

$$1. x_1 R x \wedge x_2 R x \quad \therefore \text{By def of } [x]$$

$$2. f(x) = f(x_1) = f(x_2) \quad \therefore \text{By compatibility of } f$$

$$3. f(x) = y_1 = y_2 \quad \therefore f(x_1) = y_1 \wedge f(x_2) = y_2$$

$$\therefore h_1([x]) = y_1 = y_2 = h_2([x])$$

Hence Proved!

Finally

$$\text{Supp } \alpha \in A$$

$$\text{Clearly } \alpha \in [x]$$

$$\text{As } h \text{ is a function from } A/R \rightarrow B$$

$$\exists y \in B \mid \exists x_1 \in [x] \quad (f(x_1) = y)$$

$$\text{as } x \in [x] \text{ so } f(x) = y = h([x])$$

Mergestack

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

$$(b) h: A/R \rightarrow B$$

$$\forall x \in A \quad (h([x]) = f(x))$$

$$\text{Let } x \in A \wedge y \in A \wedge x R y$$

$$\text{Then } h([x]) = f(x)$$

$$\text{Or } h([y]) = f(y)$$

$$\text{But since } x R y, \text{ so } y \in [x] \text{ so } [y] = [x] \rightarrow \text{Lemma 4-5.5}$$

$$\text{so } h([x]) = f(x) = h([y]) = f(y)$$

$$\text{so } f(x) = f(y)$$

Hence Proved!

22) By making classes & analysing, we can bring them out to be:

$$\{0, 5, 10, 15, \dots\}$$

$$\{1, 6, 11, 16, 21, 26, \dots\}$$

$$\{2, 7, 12, 17, 22, 27, \dots\}$$

$$\{3, 8, 13, 18, 23, 28, \dots\}$$

$$\{4, 9, 14, 19, 24, 29, \dots\}$$

(a) This can be simply proved by defining $f: \mathbb{Z} \rightarrow \mathbb{Z}$ that f is compatible with R . The rest of proof will follow from Exercise 21.

The proof that $F = \{(x, [x]) \mid x \in \mathbb{N} \times \mathbb{N}/R \mid f(x) = [x^2]\}$ is compatible with R is easy. Supp. $x \in \mathbb{N}$, $y \in \mathbb{N}$ & $x R y$ which means $[x] = [y]$. Then show $[x^2] = [y^2]$ is another proof given on whiteboard. Compatibility will follow from that proof.

Mergestack

Date: ____/____/____

MON - TUE - WED - THUR - FRI - SAT - SUN

Another comprehensive way is to define h

$$h = \{ ([x], [y]) \in \mathbb{N}/R \times \mathbb{N}/R \mid \exists x_1 \in [x] ([x_1]^2 = [y]) \}$$

This shows h is a function & the uniqueness of h

It can also be defined as (using R)

$$h = \{ ([x], [y]) \in \mathbb{N}/R \times \mathbb{N}/R \mid \exists x_1 \in [x] (y(x_1) = [y]) \}$$

In both cases, whole proof is similar.

(b) This is a proof by contradiction

Assume h exists

$$h: \mathbb{N}/R \rightarrow \mathbb{N}/R (\forall x \in \mathbb{N} (h([x]) = [2^x]))$$

$$\text{Then } h([0]) = [2^0] = [1]$$

$$\& h([5]) = [2^5] = [32]$$

$$[0] = [5] \text{ but } [1] \neq [32] \text{ This contradicts}$$

the fact that h is a function. ($[0] = [5]$ so this is same input but we got 2 different outputs $[1] \neq [32]$)

Some Notes about $O(x) \rightarrow Q19$

The big-O is any upper bound (e.g. $7x + 3 \in O(x)$).

Also $7x + 3 \in O(x^2)$. In Algorithm analysis, we use Θ which shows that it is a tight bound (both upper & lower bound). This shows exact growth rate. Like $O(x)$ was upper bound, Ω was lower bound. i.e

$$f \in \Omega(g) \text{ iff } \exists a \in \mathbb{Z}^+ \exists c \in \mathbb{R}^+ \forall x > a [1/c |f(x)| \geq |g(x)|]$$

$$\text{Now } f \in \Theta(g) \text{ iff } f \in O(g) \wedge f \in \Omega(g)$$

$$\text{iff } f \in O(g) \wedge g \in O(f)$$

